```
!pip install -q ultralytics

import cv2
import numpy as np
import matplotlib.pyplot as plt
from ultralytics import YOLO
from PIL import Image
```

```
model = YOLO("yolov8n.pt")

print("YOLOv8 model loaded successfully.")
```

```
YOLOv8 model loaded successfully.
```

```
!wget -q https://ultralytics.com/images/bus.jpg -O bus.jpg

image = cv2.imread("bus.jpg")
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(10, 6))
plt.imshow(image_rgb)
plt.axis("off")
plt.title("Input Image")
plt.show()
```




```
results = model("bus.jpg")

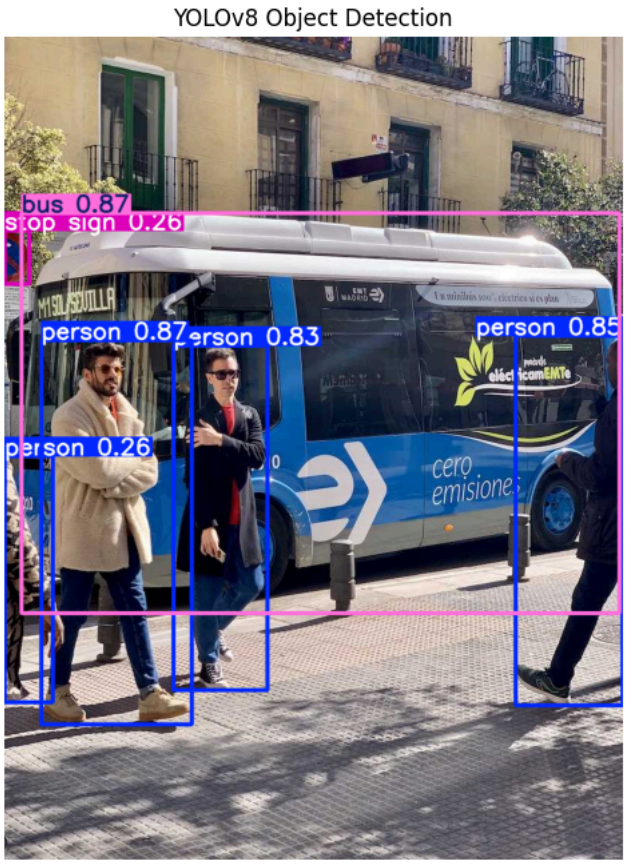
print("Object detection completed.")
```

```
image 1/1 /content/bus.jpg: 640x480 4 persons, 1 bus, 1 stop sign, 265.0ms
Speed: 7.6ms preprocess, 265.0ms inference, 25.3ms postprocess per image at shape (1, 3, 640, 480)
Object detection completed.
```

```
result = results[0]

annotated_image = result.plot()

plt.figure(figsize=(12, 8))
plt.imshow(cv2.cvtColor(annotated_image, cv2.COLOR_BGR2RGB))
plt.axis("off")
plt.title("YOLOv8 Object Detection")
plt.show()
```




```
for box in result.boxes:
    class_id = int(box.cls[0])
    confidence = float(box.conf[0])
    x1, y1, x2, y2 = map(int, box.xyxy[0])

    class_name = model.names[class_id]

    print(
        f"Object: {class_name} | "
        f"Confidence: {confidence:.2f} | "
        f"Bounding Box: ({x1}, {y1}, {x2}, {y2})"
    )
```

```
Object: bus | Confidence: 0.87 | Bounding Box: (22, 231, 805, 756)
Object: person | Confidence: 0.87 | Bounding Box: (48, 398, 245, 902)
Object: person | Confidence: 0.85 | Bounding Box: (669, 392, 809, 877)
Object: person | Confidence: 0.83 | Bounding Box: (221, 405, 344, 857)
Object: person | Confidence: 0.26 | Bounding Box: (0, 550, 63, 873)
Object: stop sign | Confidence: 0.26 | Bounding Box: (0, 254, 32, 324)
```

```
object_counts = {}

for box in result.boxes:
    class_id = int(box.cls[0])
    class_name = model.names[class_id]

    object_counts[class_name] = object_counts.get(class_name, 0) + 1

print("Detected Objects:")
for obj, count in object_counts.items():
    print(f"{obj}: {count}")
```

```
Detected Objects:
bus: 1
person: 4
stop sign: 1
```

```
import time

start_time = time.time()

results = model("bus.jpg")

end_time = time.time()
```

```
inference_time = end_time - start_time

print(f"Inference Time: {inference_time:.4f} seconds")
```

```
image 1/1 /content/bus.jpg: 640x480 4 persons, 1 bus, 1 stop sign, 137.4ms
Speed: 4.6ms preprocess, 137.4ms inference, 1.2ms postprocess per image at shape (1, 3, 640, 480)
Inference Time: 0.1572 seconds
```

```
results[0].save(filename="detected_objects.jpg")

print("Detected image saved as detected_objects.jpg")
```

```
Detected image saved as detected_objects.jpg
```

Observations:

The YOLOv8 model successfully detected multiple objects in the input image. Bounding boxes were generated around the detected objects, and each object was assigned a class label and confidence score.

The system can identify common objects such as person, bus, bicycle, car, dog, and other COCO classes, depending on what appears in the selected image.

The confidence score indicates how confident the model is about a particular detection. Higher confidence values generally indicate stronger predictions.

Questions and Answers:

Q1. Differentiate between image classification, object detection, and image segmentation. Technique Description Output Image Classification Determines what category an image belongs to. Class label Object Detection Identifies objects and determines their locations. Class labels + bounding boxes Image Segmentation Classifies individual pixels belonging to different objects or regions. Pixel-level masks

For example, if an image contains a car and a person, classification may simply say "car and person are present." Object detection identifies their locations using bounding boxes, while segmentation identifies the exact pixels belonging to the car and person.

Q2. Explain the working principle of the YOLO object detection algorithm.

YOLO stands for You Only Look Once. It is a single-stage object detection algorithm that processes an image through a neural network in one forward pass.

The image is given to the YOLO model, which predicts:

Object locations Object classes Confidence scores

The model generates bounding boxes and class probabilities for objects present in the image. Non-Maximum Suppression (NMS) is then used to remove overlapping duplicate detections.

The main advantage of YOLO is that it performs detection very quickly, making it suitable for real-time applications.

Q3. Compare YOLO, SSD, and Faster R-CNN based on speed, accuracy, and practical applications. Feature YOLO SSD Faster R-CNN Detection type Single-stage Single-stage Two-stage Speed Very high High Comparatively slower Accuracy High High Generally very high Real-time use Excellent Good Limited compared with YOLO/SSD Architecture One-pass detection One-pass detection Region proposals + classification Applications Robotics, surveillance, autonomous systems Mobile and real-time systems Detailed detection and research

YOLO and SSD are generally preferred when speed is important, while Faster R-CNN can be useful when detailed detection accuracy is more important than inference speed.

Q4. What is a bounding box? Why is it important in object detection?

A bounding box is a rectangular box drawn around a detected object.

It is generally represented using coordinates such as:

(x1, y1, x2, y2)

where the coordinates represent the position of the rectangle.

Bounding boxes are important because they provide the location and size of objects in an image.

For example, an object detector can identify a person and draw a bounding box around that person.

Q5. Explain the significance of confidence score and Intersection over Union (IoU). Confidence Score

A confidence score indicates how confident the model is that a particular detection is correct.

For example:

Person → 0.92

means that the model has a confidence score of 92% for that detection.

Intersection over Union (IoU)

IoU measures how much a predicted bounding box overlaps with the actual ground-truth bounding box.

It is calculated as:

$$IoU = \frac{Area\ of\ Intersection}{Area\ of\ Union}$$

IoU ranges from 0 to 1.

A higher IoU means that the predicted bounding box is closer to the actual object's location.

Q6. What are the advantages of using pre-trained deep learning models over traditional image processing techniques?

Pre-trained deep learning models have several advantages:

They can detect complex objects automatically. They require less training time when transfer learning is used. They can recognize many object categories. They can learn complex visual features. They generally perform better in difficult environments. They can be adapted to new datasets. Models such as YOLO can perform real-time detection.

Traditional techniques often require manually designed features and rules, whereas deep learning models learn useful features from training data.

Q7. Why are datasets such as MS COCO and Pascal VOC widely used for object detection research?

MS COCO and Pascal VOC are widely used because they provide:

Large collections of images Multiple object categories Bounding-box annotations Standardized datasets Training and testing data Common evaluation methods

They allow researchers to train and compare different object detection algorithms using standardized benchmarks.

MS COCO contains a large variety of everyday scenes and object categories, making it particularly useful for modern object detection research.

Q8. Mention five real-world applications of object detection.

Five applications are:

Autonomous vehicles – detecting vehicles, pedestrians and traffic objects. Video surveillance – detecting people and objects in security footage. Healthcare – detecting abnormalities in medical images. Retail analytics – monitoring products and customer activity. Industrial inspection – detecting defects in manufactured products.

Other applications include traffic monitoring, robotics, agriculture and smart-city systems.

Q9. What challenges are commonly encountered while detecting objects in complex real-world environments?

Common challenges include:

Poor lighting conditions. Objects being partially hidden or occluded. Objects appearing at different scales. Crowded scenes containing many objects. Motion blur. Similar-looking objects. Background clutter. Low-resolution images. Objects appearing at unusual angles. Limited computational resources for real-time detection.

These conditions can reduce detection accuracy and increase false or missed detections.

Q10. How can object detection systems be further improved using recent advancements in deep learning and computer vision?

Object detection systems can be improved through:

Larger and better-quality datasets for training. Data augmentation such as rotation, cropping and brightness changes. Transfer learning using powerful pre-trained models. Improved YOLO architectures with better feature extraction. Attention mechanisms to focus on important image regions. Better loss functions for improving bounding-box prediction. Model optimization and quantization for faster inference. GPU acceleration for real-time applications. Multi-scale detection for objects of different sizes. Combining detection with tracking for video applications.

Recent developments in transformer-based vision models and improved detection architectures can also improve performance on challenging datasets.